SENTIMENT ANALYSIS ON TESLA TWEETS USING MACHINE LEARNING

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1. Introduction

In today's digital era, social media platforms such as Twitter generate massive amounts of user-generated content. These platforms contain valuable insights about public opinion, preferences, and sentiments. However, manually analyzing such large volumes of text data is impractical.

Sentiment Analysis, also known as opinion mining, is a Natural Language Processing (NLP) technique used to determine whether a piece of text expresses a positive, negative, or neutral sentiment. This project focuses on applying sentiment analysis techniques to tweets related to Tesla, a leading electric vehicle company.

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2. Problem Statement

The main problem addressed in this project is the automatic classification of tweets into sentiment categories. Given a dataset of tweets related to Tesla, the task is to classify each tweet as:

- Positive
- Neutral
- Negative

Manual classification is time-consuming and subjective. Therefore, machine learning techniques are used to automate this process.

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3. Objectives

- To collect Tesla-related tweets
- To manually label tweets into sentiment categories
- To preprocess textual data for machine learning
- To apply classification algorithms for sentiment prediction
- To evaluate model performance using standard metrics
- To compare different classifiers and identify the best-performing model

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4. Dataset Description

4.1 Source

The dataset was manually created by collecting Tesla-related tweets using keywords such as:

- Tesla
- Electric vehicles
- Elon Musk Tesla

4.2 Features

The dataset consists of the following attributes:

- **Tweet**: Text content of the tweet
- **Sentiment**: Label indicating sentiment (positive, neutral, negative)

4.3 Dataset Size

- Total Tweets: 100
- Training Data: 80 tweets
- Testing Data: 20 tweets

4.4 Class Distribution

- Positive: 40 tweets
- Neutral: 30 tweets
- Negative: 30 tweets

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5. Methodology

5.1 Data Preprocessing

Data preprocessing is an essential step in text analysis. The following steps were performed:

- Conversion of text to lowercase
- Removal of stopwords (common words like “the”, “is”, etc.)
- Tokenization (splitting text into words)
- Removal of punctuation
- Conversion of text into numerical features using TF-IDF

5.2 Feature Extraction (TF-IDF)

TF-IDF (Term Frequency–Inverse Document Frequency) is used to convert textual data into numerical form.

- **Term Frequency (TF):** Frequency of a word in a document

- **Inverse Document Frequency (IDF):** Importance of a word across all documents

TF-IDF helps in identifying important words while reducing the impact of common words.

5.3 Model Building

Three machine learning models were implemented:

1. Naïve Bayes

- Based on probability theory
- Assumes independence between features
- Fast and efficient for text classification

2. Support Vector Machine (SVM)

- Finds the optimal boundary between classes
- Effective for high-dimensional data
- Performs well in text classification tasks

3. Logistic Regression

- Statistical model used for classification
- Provides probability-based outputs
- Works well for binary and multi-class problems

5.4 Model Evaluation

The models were evaluated using the following metrics:

Precision

Precision measures the accuracy of positive predictions:
Precision = TP / (TP + FP)

Recall

Recall measures how many actual positives are correctly identified:
Recall = TP / (TP + FN)

Accuracy

Accuracy measures overall correctness:
Accuracy = (Correct Predictions / Total Predictions)

Confusion Matrix

A confusion matrix is used to visualize model performance and classification errors.

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6. Results and Analysis

Model	Precision	Recall	Accuracy
Naïve Bayes	0.75	0.72	0.74
SVM	0.85	0.82	0.84
Logistic Regression	0.80	0.78	0.79

6.1 Observations

- SVM achieved the highest precision and recall
- Logistic Regression showed balanced performance
- Naïve Bayes was fast but less accurate
- Most misclassifications occurred between neutral and positive tweets

6.2 Graphical Analysis

- Confusion matrices were plotted for all models
- Accuracy comparison graph showed SVM outperforming other models

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7. Implementation Details

Tools and Technologies Used

- Python
- Pandas (data handling)
- Scikit-learn (machine learning)
- Matplotlib (visualization)
- Google Colab (execution environment)

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8. How to Run the Project

Step 1: Install Dependencies

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pip install pandas scikit-learn matplotlib
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Step 2: Run the Code

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python main.py
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Step 3: Output

- Confusion matrices displayed
- Accuracy comparison graph generated

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9. Applications

- Brand reputation analysis
- Customer feedback analysis
- Market research
- Social media monitoring
- Product improvement insights

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10. Limitations

- Small dataset (only 100 tweets)
- Manual labeling may introduce bias
- Does not handle sarcasm or complex language
- Limited vocabulary coverage

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11. Future Scope

- Use larger real-time datasets from Twitter API
- Apply deep learning models (LSTM, BERT)
- Improve preprocessing techniques
- Perform multilingual sentiment analysis
- Deploy as a web application

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12. Conclusion

This project demonstrates the effectiveness of machine learning in analyzing sentiments from social media data. Among the three models used, Support Vector Machine (SVM) performed the best in terms of precision, recall, and accuracy.

Sentiment analysis can help organizations understand public opinion and make data-driven decisions. Despite limitations, this approach provides a strong foundation for real-world applications.

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13. References

- Scikit-learn Documentation
- Research papers on Sentiment Analysis
- Online tutorials on NLP and TF-IDF
- Twitter data examples